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SCIENCE AND TECHNOLOGY

INTRODUCTION

Science and Technology are fundamentally altering the way people live, connect, communicate and transact, with profound effects on economic development. Science and Technology are key driver's scientific revolutions and their embracement by society led to economic advances, improvements in health systems, education and infrastructure.

The technological revolutions of the 21st century is emerging from entirely new based sectors, based on microprocessors, telecommunications, biotechnology and nanotechnology. Products are transforms business practices across the economy, as well as the lives of all who have access to their effects. The most remarkable breakthroughs will come from the interaction of insights and applications arising when these technologies converge.

Access and application are critical. Service and technology are the differentiators between countries that can tackle poverty effectively by growing and developing their economies and those that are not.

The extent to which developing economies emerge as economic powerhouses depend on their ability to grasp and apply insights from science and technology and use them creatively. Innovation is the primary driver of technological growth and drives higher living standards.

In India:

In 1947, with the emergence of a new politically independent nation, India continued to march ahead pursuing a programme of using modern science and technology for national development. Today India spends about 1.5 percent of its GNP on science and technology. In this effort not only has India established capabilities of its own but has

also cooperated with developed as well as developing countries in its progress towards the use of science and technology for national development.

Soon after Pandit Jawaharlal Nehru became the Prime Minister of India, he created a Ministry of Scientific Research and Natural Resources, and actively supported the atomic energy programme for peaceful purposes.

In 1948, the Atomic Energy Act was passed and the Department of Atomic Energy was directly under his charge. Till his death in 1964, he was the Chairman of the Council of Scientific and Industrial Research. His long association with the India Science Congress Association is well known. Under the farsighted leadership of Nehru, the nation, the government and the public leaders became committed to the promotion of science and technology for national development in a phased manner.

The enthusiastic efforts of Mr. Shanti Swarup Bhatnagar led to the expansion of the Council of Scientific and Industrial Research into a chain of national laboratories spanning a wide spectrum of science, technology, engineering and biomedical sciences. The vision of Homi .J. Bhabha led to advanced research in nuclear energy and other fundamental areas through the creation of the Tata Institute of Fundamental Research (TIFR) and what has now come to be known as the Bhabha Atomic Research Center (BARC), and the entire gamut of activities today coming under the Atomic Energy Commission.

Soon after assuming office, Nehru appointed a Scientific Man-power Committee and had the satisfaction of seeing five Institutes of Technology come up at Kharagpur, Bombay, Madras, Kanpur and Delhi, besides a number of regional engineering colleges. A number of institutions for specialized training such as the National Institute of Foundry and Forge Technology, School of Planning and Architecture, the Institutes of Management and the All India Institute of Medical Sciences were set up. A similar expansion took place in science education.

The number of universities and science graduates and post-graduates multiplied. Nehru diversified the area of operation in science and technology. India was the first country, originally on the foot-pound system, to change over the metric system during the present century.

In 1948, Nehru directed the SCIR to prepare a National Register of Scientific and Technical Personnel. The Defence Science Organisation was set up in 1948, on the advice of Professor P.M.S. Blackett, for the scientific evaluation of weapons and equipment, operational research and special studies using scientific technique. Prime Minister, Indira Gandhi gave the highest priority to self- reliance in science and technology and the achievement of self-sufficiency in food.

In 1971, recognizing the importance of developing integrated and self-reliant electronic capabilities in the country, she set up the Electronics Commission. There have been many accomplishments in the field. To ensure that developmental activities took place in harmony with the environment, Mrs. Gandhi created a new Department of Environment at the Centre in 1980.

It was at her initiative that the first Indian scientific expedition to Antarctica took place in December 1981. She was deeply aware of the great importance of energy for development and, in particular, the pressing needs in rural areas. Accordingly, she set up a Commission on Additional Sources of Energy in March 1981, and thereafter a Department of Non-Conventional Energy Source.

India's development plans have consistently emphasised the need for sustained investment in research and related activities leading to creation of substantial capacity and capabilities in science and technology (S&T). The fruits of this effort are evident in India's nuclear and space programmes, information and communication technology services, automotive and pharmaceuticals industries and other areas.

As the Indian economy continues on the path of rapid, more inclusive and sustainable growth, it will be necessary to ensure that India's capabilities in S&T grow in strength. This is especially important if India is to become one of the major economies of the world over the next 20 years.

The country needs to move up from investing 1 per cent of gross domestic product (GDP) in the R&D sector to 2 per cent of GDP and more, as has been the case with several developed and emerging economies for quite some time now. This must be

achieved not only through an additional government effort, but also a much-increased private sector effort.

ROLE OF THE SCIENCE AND TECHNOLOGY

Introduction

About 200 years ago the pace of technological change in western society began to quicken. Wind, water, and animal power, with their limitations of place and capacity, were supplemented and then replaced by the steam engine, which went on to power the factories of the industrial revolution. The railroad made it possible to move things and people quickly over great distances. The telegraph and, later, the telephone carried communications across the countryside. Electric lighting supplanted the dim glow of candles, kerosene, and gas lights.

By the beginning of the twentieth century, the notion of progress was closely linked with technological development, and that linkage intensified in the following decades. The automobile and the airplane changed not only travel but the nature of our cities and towns. Radio and then television brought more of the outside world into everyone's homes. Knowledge about the causes of diseases brought new treatments and preventive measures. Computers appeared, and soon the transistor made them smaller, more powerful, more accessible, and cheaper.

Today, the system by which research and development leads to new products is fundamentally different than it was in the nineteenth century. To the role of the individual inventor has been added the power of organized scientific research and technological innovation. Organized research and development, which are increasingly international in character, have greatly increased the production of new knowledge. Deeper understanding of living organisms is leading toward cures of diseases once thought untreatable.

Basic insights in materials science enable the development of structures that are lighter, stronger, and more durable than anything available before. The computer and novel modes of communication, such as optical fibers, bring new, interactive modes of

work and more capable machinery. These new devices and new ways of working, in turn, speed the growth and dissemination of new knowledge.

The accumulation of scientific knowledge and new technologies has transformed human life. Technologies have helped provide many—though far from all—people with standards of warmth, cleanliness, nutrition, medical care, transportation, and entertainment far beyond those of even the wealthy two centuries ago. They have also presented us with difficult questions about how to use science and technology most effectively to meet human needs.

The rapid rate of material progress can continue, but it is not inevitable. The extent to which the products of science and technology are useful depends on the needs of society. Each of the four areas discussed in this chapter—industrial performance, health care, national security, and environmental protection—uses these products in different ways. Progress is more likely if we understand these differences. Only then can we effectively translate scientific and technical understanding into the techniques, tools, and insights that improve the quality of our lives.

IN INDUSTRY:

Industries differ in the manner and extent to which they use the results of research. Some, such as the semiconductor industry, the biotechnology industry, and parts of the chemical industry, were created and shaped almost entirely by ideas that grew out of science. The technologies at the heart of these industries were initially characterized more by promise than by real products.

Semiconductors were in this stage right after the invention of the transistor; more recently, biotechnology went through this stage after the development of recombinant DNA techniques. High-temperature superconductivity is a scientific discovery that shows promise of leading to new industries and is in this stage today.

As science-based industries continue to develop, they remain closely dependent on continuous inputs of new science, often produced by university researchers. These industries depend as well on the technological development of these ideas in order to grow and to widen their range of products.

At an early stage, these industries tend to be small, to move at a fast technical and competitive pace, and to have enormous potential. Biotechnology is now in this stage.

In a more mature stage, a science-based industry may still be growing quickly, but it depends on the progress of academic scientists. The semiconductor industry, for example, moves at a fast-technical pace and requires increasingly detailed knowledge of its materials and, as the individual transistors buried in its chips become ever smaller, even of new quantum phenomena. But its scientific needs are met almost entirely by the work of semiconductor scientists and engineers working in the plants and laboratories of the semiconductor companies. Indeed, industry scientists are often the only ones with the detailed knowledge needed to make incremental improvements in the technologies.

Another example of an industry at a mature stage is the aircraft industry, where thousands of scientists and engineers are required to deal with the enormous complexities of new plane design. Investments in manufacturing tools and plants are often measured in hundreds of millions of dollars. Only major companies can act on this scale, and only they have the technological knowledge and experience needed to design these complex products.

The most mature industries—for example, the automobile or construction industries—move at a slower technological pace and require fewer inputs from current science, whether generated by their own laboratories or by university research. Many of these were not based on science even at their birth. They do, however, require the highest levels of technological and production know-how.

For industries that rely on high technology but are technically self-contained (such as the semiconductor industry) and industries that do not depend heavily on current science (such as the automobile industry), the results of current fundamental research are generally not decisive. Japan, which has not been a leading research power, has exhibited great strength in such industries.

In these areas, productivity gains and product leadership can be attained by a number of strategies largely separate from scientific research but highly dependent on engineering, such as developing new technology in corporate laboratories, improving

the development cycle to hasten the marketing of improved products, better coordination of design and manufacture, maximizing the creative capabilities of employees, and responding quickly to changes in consumer preferences. Additional university research can help, but it will be of peripheral importance to such industries. Nor can research rescue a failing industry that has difficulties in other areas.

ROLE OF SCIENCE AND TECHNOLOGY IN MEETING OTHER NATIONAL OBJECTIVES:

In addition to their influence on industrial performance, science and technology are directly involved in efforts to achieve a number of other important national goals. As in the case of industry, many other factors must also be in place for the goals to be achieved, but science and technology provide many of the crucial insights and techniques that enable progress.

The following sections briefly describe some of the linkages between science and technology and several of these goals.

Health Care:

Maintenance of health and prevention of illness are among the highest goals of our society. Science and Technology have become critical factors in achieving those goals, and the health sciences—including the life sciences, health services research, and public health research—will remain vital elements in the promotion of the nation's wellbeing. In health care, as in other areas, science and technology are embedded in much broader social and institutional structures. For example, a research discovery can lead to experimental products in a very short time.

Yet those products may require very long lead times to bring to market because of the need to ensure their safety and efficacy. The most visible public policy issue in health care today is cost. Many of the medical products generated by research and development, such as vaccines, actually reduce total health care costs.

Other new products derived from research and development, such as complex imaging devices and expensive surgical procedures, raise costs in the short term while

enhancing overall care. Still other procedures reduce unit treatment costs, but these reductions make treatments more available and thus increase demand and total costs.

The development and pricing of health care products are unusual for a number of reasons. In a normal market economy, differences in the costs of technologies are reflected in the level of use. But our current system of health care reimbursement insulates patients from the true costs. In addition, the government directly regulates many aspects of medical technology to ensure safety and control costs, further distorting market signals. Finally, health care involves such basic human conditions as birth, disease, and ultimately death. Under such conditions, individual consumers often ignore economic considerations; yet the total cost of health care is a matter of enormous national concern.

The effects of technical progress on costs depend to a large extent on the social and institutional structures surrounding the health care system. As the nation undertakes a broad reassessment of its health care system, a central challenge is to create administrative structures that promote the development of medical technology while improving care and containing costs.

National Security

Since World War II, the United States has sought military advantage through technological rather than numerical superiority. For example, technological superiority in the hands of a well-trained military contributed greatly to the success of the Persian Gulf War.

The United States will continue to rely on this strategy to retain military advantage, but the sources of new military technology are shifting. In the past, the segment of industry that has supplied both hardware and software to the U.S. military has been largely separate from civilian industry. This segment of industry has had essentially one customer, and its requirements were focused on product performance more strongly than on cost.

In the 1950s and 1960s, the defense industry produced much technology of value to civilian industry. But today the technological sophistication of civilian industry in many cases surpasses that of the defense industry. As a result, the military has become more dependent on civilian technologies. This trend will make improvements in national security more dependent on overall national economic performance.

A major challenge facing the military today is to maintain technological superiority in the face of declining defense budgets. Meeting this challenge will require a reexamination of the broad scientific and technological base that contributes to military needs, including research and development in government laboratories, in industry, and in universities.

Environmental Protection

Over the past two decades, the United States has recognized and has made substantial progress in curbing the degradation of the environment. Nevertheless, difficult problems remain. Environmental degradation continues to accompany many aspects of economic growth. Emissions and effluents of contaminated materials continue, waste disposal plagues urban areas, forests continue to be devastated, and biodiversity losses are growing. At the same time, science and technology have exposed new issues of great complexity and uncertain consequences, such as global warming, acid precipitation, the destruction of the stratospheric ozone layer, and the contamination of water supplies.

By the middle of the twenty-first century, the human population is projected to double to around 11 billion people, and, to meet their basic needs, the global economy will need to be several times larger than it is now. Many industrial and agricultural practices and products used today in energy and food production, transportation, and manufacturing will need to be restructured to prevent pollution if sustainable economic growth is to be achieved. In some situations, existing technologies can be made cleaner and more efficient; in others, entirely new technologies, including energy technologies, will be needed.

Almost all fields of science and technology can contribute to the reduction of environmental degradation. Biotechnology, materials science and engineering, and information technologies can enable the efficient use of raw materials and prevent pollution at the source. Reducing and preventing pollution is an important goal of the new field of industrial ecology, which, by examining industrial processes, strives to maintain sustainable technological growth.

ACHIEVEMENTS OF INDIA IN SCIENCE AND TECHNOLOGY

INTRODUCTION

India was a land of sages and seers as well as a land of scholars and scientists. Research had shown that from making the best steel in the world to teaching the world to count, India was actively contributing to the field of science and technology centuries long before modern laboratories were set up. Many theories and techniques discovered by the ancient Indians have created and strengthened the fundamentals of modern science and technology.

India has the second largest group of scientists and engineers in the world. In the context of technological advancements and scientific achievements, Indian scientists have developed several projects and augmented the life of world populace. Scientific research conducted by Indian has transformed the way the world works, from healing and eradicating lethal diseases to understanding the world. While some of these ground breaking contributions have been acknowledged, some are still unknown to most. Some of the major ancient breakthrough achievements by Indians in different fields are discussed below.

Mathematics

1. **The Idea of Zero** – Mathematician **Aryabhata** was the first person to create a symbol for zero and it was through his efforts that mathematical operations like addition and subtraction started using the digit, zero. The concept of zero and its integration into the place-value system also enabled one to write numbers, no matter how large, by using only ten symbols. (The mention of Zero is also found in **Bakhshali Manuscript)**

2. **Bakshali manuscript** is a mathematical text written on birch bark that was found in 1881 in the village of Bakhshali, (near Peshawar in present-day Pakistan) dated A.D. 224-383. It is notable for being **'the oldest manuscript in Indian mathematics".** In 2017, three samples from the manuscript were shown by radiocarbon dating.
3. The heliocentric Theory – Mathematicians of ancient India often applied their **mathematical knowledge to make accurate astronomical predictions.** The most significant among them was Aryabhata whose book, **Aryabhatiya** represented the pinnacle of astronomical knowledge all the time. He correctly propounded that the Earth is round, rotates on its axis and revolves around the Sun i.e. the heliocentric theory. He also made predictions about the solar and lunar eclipses, duration of the day as well as the distance between the Earth and the Moon.
4. **The Invention of Binary Number –** Vedic Indian Scholar **Pingala Invented Binary Number System** in 2nd century BCE in his book **Chandahsastra.** Pingala (in Chandahsastra) has assigned the following combinations of **zero** and **one** to represent various numbers, much in the same way as the present-day computer programming procedures.
5. **The decimal system** – India gave the ingenious method of expressing all numbers using ten symbols – the decimal system. In this system, each symbol received a value of position as well as an absolute value. Due to the simplicity of the decimal notation, which facilitated calculation, this system made the uses of arithmetic in practical inventions much faster and easier.
6. **Fibonacci Number** – The Fibonacci numbers and their sequence first appear in Indian mathematics a matrameru mentioned by Pingala in connection with the Sanskrit tradition of prosody.
7. **Brahmagupta-Fibonacci identify expresses the product of two sums of two squares as a sum of two squares in two different ways.** Hence the set of all sums of two squares is closed under multiplication, later on, the methods for the formation of these numbers were given by mathematicians Virahanka, Gopala and Hemacandra, much before the Italian mathematician

Fibonacci introduced the fascinating sequence to Western European mathematics.

8. **Algebraic abbreviations** – The **mathematician Brahmagupta** had begun using abbreviations for unknowns by the 7th century. He employed abbreviations for multiple unknowns occurring in one complex problem. Brahmagupta also used abbreviations for square roots and cube roots.
   a. **Brahmagupta's theorem** states that if a cyclic quadrilateral is an Orth diagonal (that is, has perpendicular diagonals), then the perpendicular to a side from the point of intersection of the diagonals always bisects the opposite side. It is named after the Indian mathematician Brahmagupta.
9. **Chakravala method** – The Chakravala method, a cyclic algorithm to solve indeterminate quadratic equations is commonly attributed to **Bhaskara II,** (c. 1114 – 1185 CE) although some attribute it to **Jayadeva** (950-1000 CE).
   a. **Jayadeva's** method was later refined by Bhaskara II in his **Bijaganita** treatise to be known as the Chakravala method, Chakra meaning "wheel" in Sanskrit, relevant to the cyclic nature of the algorithm. Concerning the **Chakravala method,** E.O. Selenuis held that no Eruropean performances at the time of Bhaskara, nor much later, came up to its marvellous height of mathematical complexity.
10. **Astronomy:** Ancient India's constitutions in the field of astronomy are well known and well documented. The earliest references to **astronomy are found in the Rig Veda,** which is dated 2000 BC. During the next 2500 years, by 500 AD, ancient Indian astronomy has emerged as an important part of Indian studies, and its effect is also seen in several freatises of that period. In some instances, astronomical principles were borrowed to explain, matters, about astrology, like the casting of a horoscope. Apart from this linkage of astronomy with astrology in ancient India the science of astronomy continued to develop independently, and culminated into original findings, like.
    1. The calculation of **occurrences of eclipses.**
    2. Determination of **Earth's circumference.**
    3. **Theorizing about the theory of gravitation.**

4. Determing the Sun was a star determination of some **planets** under our solar system.

11. The **Plelades** hold a prominent place as the mothers or wet nurses of the newborn infant in one of the most ancient and central Hindu myths, that of the birth of the war-god Rudra/Skanda, who represents, among other things, the victorious rising Sun (and as the vernal Sun the new year). The Pleiades are said to have been the wives of the seven sages, who are identified with the seven stars of the Great Bear.
12. The **Great Bear's Old Tamil name elu-mean 'seven-star'** corresponds to the combination of the pictograms '7' + 'fish', which alone constitutes the entire text of one finely carved Indus seal. The **Satapatha-Brahmana states that the six Pleiades were separated** from their husbands on account of their infidelity; other texts specify that only one of the seven wives. Arundhati, remained faithful and was allowed to stay with her husband; she is the small star Alcor in the Great Bear, pointed out as a paradigm of marital virtue to the bride in the Vedic marriage ceremonies.
13. Evidence for the Harappan origin of this myth is provided, among other things, but Indus seals which show a row of six or seven human figures; their female character is suggested by the one long plait of hair, which to the present day has remained characteristic of the Indian ladies.

Physics

1. From ancient times, Indian philosophers believed that except ether or space, all other elements were physically palpable and hence comprised of small and minuscule particles of matter. They believed that the smallest particle which could not be subdivided further was **paramanu, a** Sanskrit word. Paramanu is made of two Sanskrit words, **param meaning** ultimate and meaning atom. Thus, the term "paramanu" literally means 'beyond atom', and this was a concept at an abstract level which indicated the possibility of splitting the atom, which is row the source of atomic energy. The term 'atom' however should not be conflated with the concept of the atom as it is understood today.

2. **Kanad's Theory of Atom** – Ancient Indian Sage and Philosopher, Kanad (happen lived between 6th century and 2nd century BCE) was also known as **Kashyap, Kanada,** who founded the **Vaisheshika** (one of the six orthodox schools of Vedic Systems) schools of Indian philosophy invented theory of atom which is an indestructible particle of matter also. He used invariance arguments to determine the properties of the atoms.

Medicine

1. **Charaka** authored a foundational text, **Charakasamhita,** on the ancient science of **Ayurveda.** Referred to as the **Father of Indian Medicine,** Charaka was the first physician to present the **concept of digestion, metabolism and immunity** in his book. Charaka's **ancient manual on preventive medicine** remained a standard work on the subject for two millennia and was translated into many foreign languages, including Arabic and Latin.
2. **Ayurvedic and Siddha medicine** – Ayurveda and Siddha are ancient systems of medicine practised in South Asia. Ayurvedic ideas can be found in the Hindu. Text (mid-first millennium BCE), The Sanskrit word **ayur-vedah 'knowledge (Veda) for longevity (ayur)'.** Siddha medicine is most prevalent in South India, and is transmitted in tamil, not Sanskrit, texts. **Herbs and minerals are basic raw materials of the Siddha therapeutic system** whose origins may be dated to the early centuries CE.
3. **Cataract surgery** – The first cataract surgery is said to have been performed by the ancient Indian physician **Sushruta,** way back in 6th century BCE. To remove the cataract from the eyes, he used a curved needle **Jabamukhi Salaka,** to loosen the lens and push the cataract out of the field of vision. The eye would then be bandaged for a few days till it healed completely. Sushruta's surgical works were later translated to Arabic language, and through the Arabs, his works were introduced to the West.
4. **Plastic Surgery** – The earliest mention of plastic surgery was found in **Sushruta Samhita** Written by Sushruta in 6th Century BC. Sushruta Samhita is considered to be one of the most comprehensive textbooks on ancient surgery.

The text mentions various illnesses, plants, preparations and cures along with complex techniques of plastic surgery. The Sushruta Samhita's most well-known contribution to plastic surgery is the reconstruction of the nose, also known as rhinoplasty.

5. **Vagbhata** is one of the most influential classical writers of Ayurveda. Several works are associated with his name as author, principally the **Ashtangasangraha** ("Compendium of Medicine") and the **Ashtangahridayasamphita** ("Heart of Medicine"). Both works make frequent reference to the earlier classical works, the Charaka Samhita and the Sushruta Samhita. Vagbhata is said, in the closing verses of the Ashtangasangraha, to have lived in Sind (today in Pakistan). In his writings praise for the Shiva has been shown by name at the start of the Ashtangasargraha, and his praise of the Shiva under the title "Unprecedented Teacher" in the opening verse of the Ashtangahridayasamhita.

Metallurgy

1. **Wootz steel** - A pioneering steel alloy matrix developed in India, Wootz steel is **crucible steel,** characterized by a pattern of bands that was known in the ancient world by many different names such as *Ukku, Hindiwani, and Seric Iron*. This steel was used to make the famed Damascus swords of yore that could cleave a free-falling silk scart or a block of wood with the same ease. Produced by the Tamils of the Chera Dynasty, this finest steel of the ancient world was made by heating black magnetite ore in the presence of carbon in a sealed clay crucible kept inside a charcoal furnace.
2. **Iron pillar of Delhi** – The iron pillar of Delhi is a 7 m (23 ft) **Kirtistambha (victory column),** originally erected and dedicated as dhvaja (banner) to Hindu deity Lord Vishnu in 3rd to 4th century CE by king Chandragupta II. The pillar has attracted the attention of archaeologists and materials scientists because of its high resistance to corrosion.
3. **Smelting of Zinc** – India was the first to smelt zinc by the distillation process, an advanced technique derived from a long experience of ancient alchemy. The ancient Persians had also attempted to reduce zinc oxide in an open furnace but

had failed. Zawar in the Tiri Valley of Rajasthan is the world's first known ancient zinc smelting site. The distillation technique of zinc production goes back to the 12th Century AD and is an important contribution of India to the world of science.

4. **Seamless metal globe –** Considered one of the most remarkable feats in metallurgy, the first seamless celestial globe was made in Kashmir by Ali Kashmiri ibn Luqman in the reign of Emperor Akbar. In a major feat in metallurgy, Mughal metallurgists pioneered the method of lost-wax casting to make twenty other globe masterpieces in the reign of the Mughal Empire. Before these globes were rediscovered in the 1980s, modern metallurgists believed that it was technically impossible to produce metal globes without any seams, even with modern technology.
5. **Iron cased Rocket –** The first iron-cased rockets were developed in the 1780s by Tipu Sultan of Mysore who successfully used these rockets against the larger forces of the British Eash India Company during the Anglo-Mysore Wars. He crafted long Iron tubes, filled them with gunpower and fastened them to bamboo poles to create the predecessor of the modern rocket. With a range of about 2 km, these rockets were the best in the world at that time and caused as much fear and confusion as damage. Due to them, the British suffered one of their worst ever defeats in India at the hands of Tipu.

DEVELOPMENT OF SCIENCE AND TECHNOLOGY

Science & Technology (S & T) as an integral part of the Indian culture has been enabling force behind the longevity of India's civilisation. It not only allowed to question natural landscape with a sense of enquiry but also opened the gate to prosperity by helping society to lead technologically driven life.

Indus Valley Civilisation

The archaeological remains at various sites of the Indus Valley Civilization allow us to know about its technological progress. The key scientific developments and achievements by Indus Valley people include:

1. **Architecture and Civil Engineering** – The well planned urban centres based on definite layout pattern with scientific roads; drainage system (with use of corbel technique), public structures (like granaries and great bath); world's first tidal port at the head of Gulf of Cambay in Gujarat were much ahead of times and precursor to the modern concept of architecture and civil engineering.
2. **Irrigation System** – They had an innovative irrigation system of artificial reservoirs that led them to achieve size and prosperity.
3. **Weights, Measurement and Metallurgy** – There were standardised systems of weights and measurements along with calibration in multiple subdivisions.
4. **Medical Science** – It was announced in Journal Nature in April 2006 that oldest evidence in human history for the drilling of tooth in vivo, that is, in a living person, was found in Mehrgarh around 7000 BC. The people of Indus Valley Civilization practised Trephination which is kind of medical intervention making a burr hole in the skull to treat migraines and mental disorders.

Post-Indus Valley

During this time India was a land of sages, saints and seers as well as a land of scholars and scientists. Ancient India's contribution to science and technology include:

1. **Mathematics** – Vedic literature is replete with concepts of zero, the techniques of algebra and algorithm, square root and cube root. Arguably, the origins of Calculus lie in India 300 years before Leibnitz and Newton.
2. **Astronomy** – Rig Veda (2000 BC) refers to astronomy.
3. **Physics** – Concepts of atom and theory of relativity were explicitly stated by an Indian Philosophies around 600 BC.
4. **Chemistry** – Principles of chemistry did not remain abstract but also found expression in the distillation of perfumes, aromatic liquids, manufacturing of dyes and pigments, and extraction of sugar.
5. **Medical Science & Surgery** – Around 800 BC, the first compendium on medicine and surgery complied in ancient India.
6. **Fine Arts** – Vedas were recited, and recitation has to be correct, which gave rise to a finer study of sound and phonetics. The natural corollary was the emergence of music and other forms of performing arts.

7. **Mechanical & production technology** – Greek historians, have testified to smelting of certain metals in India in the 4th century BC.
8. **Shipbuilding & navigation** – Sanskrit and Pali texts have several references to maritime activity by ancient Indians.

Medieval India

During the medieval period, Science and Technology in India developed two facets: one concerned with the already chartered course of earlier traditions and other with the new influences which came up as a result of Islamic and European impact.

Mathematics:

1. Narayana Pandit, son of Narasimha Daivajna, was well known for his works in Mathematics – Ganitakaumudi and Bijaganitavatamsa.
2. Gangadhara, in Gujarat, wrote Lilavati Karamdipika, Suddhantadipika, and Lilavati Vyakhya. These works provided rules for trigonometrical terms like sine, cosine tangent and cotangent.
3. Nilakantha Jyotirvida complied Tajik, introducing a large number of Persian technical terms.
4. Faizi, at the behest of Akbar, translated Bhaskara's Bijaganit. Akbar ordered to make Mathematics as a subject of study, among others in the education system.

Biology:

1. Hamsadeva complied a work in the field of Biology entitled Mrga-paksi-sastra in the thirteenth century. It gives a general, though not always scientific, an account of some animals and birds of hunting.
2. Both Babur and Akbar, in spite of being busy in their political preoccupations and war, found time to study the work. 'Akbar had a special interest in producing good breeds of domestic animals like elephants and horses.
3. Jahangir, in his work Tuzuk-i-Jahangiri, recorded his observations and experiments on breeding and hybridisation. He described 36 species of animals.

Chemistry:

1. An important application of Chemistry was in the production of paper. Kashmir, Sialkot, Zafarabad, Patna, Murshidabad, Ahmedabad, Aurangabad and Mysore became well-known centres of paper production.
2. The Mughals knew the technique of production of gunpowder and its use in gunnery, another application of Chemistry. The Indian crafts persons learnt the technique in evolved suitable explosive composition. The work Sukraniti attributed to Sukracarya contains a description of how gunpowder can be prepared using saltpetre, sulphur and charcoal in different ratios for use in different types of guns.
3. The work Ain-i-akbari speaks of the regulation of the perfume office of Akbar.

Astronomy

Astronomy was another field that flourished during this period, in astronomy, some commentaries dealing with the already established astronomical notions appeared.

1. Mahendra Suri, a court astronomer of Emperor Firoz Shah, developed an astronomical, instrument 'Yantraja'.
2. Maharaja Sawai Jai Singh-II of Jaipur was a patron of Astronomy. He set up the five astronomical observatories in Delhi, Ujjain, Varanasi, Mathura and Jaipur.

Medicine:

The Ayurveda system of medicine did not progress as vigorously as it did in the ancient period because of the lack of royal patronage. However, some important treatises on Ayurveda like the Sarangdhara Samhita and Chikitsasamgraha by Vangasena, the Yagaratbajara and the Bhavaprakasa of Bhavamisra were compiled.

1. The Sarangdhara Samhita, written in the thirteenth century, includes the use of opium in its material medical and urine examination for diagnostic purpose. The drugs mentioned include metallic preparation of the rasachikitsa system.
2. The Unani Tibb system of medicine flourished in India during the medieval period. Ali-bin-Rabban summarised the whole system of Greek medicine as well as the Indian medical knowledge in the book, Firdausu-Hikmat.

Colonial India

The continuous march towards scientific advancement paralleled colonial age of frequent struggle for power, resources and ideology. According to the account of Stephen Oliver Fought & John F. Guilmartin, in 18th century. Hyder Ali, prince of Mysore, developed war rockets with an important change: the use of metal cylinders to contain the combustion powder. Although the hammered soft iron he used was crude, the bursting strength of the container of black powder was much higher than the earlier paper construction.

Thus, a greater internal pressure was possible, with a resultant greater thrust of the propulsive jet. Tippu Sultan, continued to develop and expand the use of rocket weapons, reportedly increasing the number of rocket troops from 1,200 to a corps of 5,000.

In the 19th century, The British education system, aimed at producing able civil and administrative services candidates, exposed some Indian scientific scholars to a foreign institution. Sir Jagadish Chandra Bose, Prafulla Chandra Ray, Satyendra Nath Bose, Meghnad Saha, P.C. Mahalanobis, Sir C.V. Raman were among the notable among them.

Post-Independence

The innate ability to perform creatively in science came to be backed with an institutional set-up and strong state support after the country's independence in 1947. As the economic development of the country at that time had not reached a stage where the private sector could play a leading role in the S&T development, the Government took this task upon itself and provided the initial momentum.

India's first Prime Minister Pandit Jawaharlal Nehru gave whole-hearted support to a concerted programme for the promotion of S&T in the country. Nehru firmly believed that Science and Technology could be the twin tools that would help bring about social equity and economic development to enable India to join the mainstream of the world community.

This conviction was reflected in the Scientific Policy Resolution (SPR) of 1958, the aim of which was "to foster, promote and sustain the cultivation of sciences and scientific research in the country and to secure for the people all the benefits that can accrue from the acquisition and application of scientific knowledge". This has resulted in a broad-based and extensive S&T network. As a result, many new S&T departments and laboratories were set up, and the pursuance of scientific research started in an organized manner.

The departments of Atomic Energy, Science and Technology, and Space were thus among the first S&T departments in the country with the Prime Minister, himself taking the reins. The Industrialisation of the country, according to the founding fathers of modern India, was essential for improving the quality of life. Consequently, the Council of Scientific and Industrial Research (CSIR) expanded with the establishment of various institutions dedicated to the development of specific industries.

SCIENCE AND TECHNOLOGY DURING FIVE YEAR PLANS

The five-year planning on science and technology aimed to foster, promote and sustain the cultivation of the sciences and scientific research and to encourage individual for dissemination of scientific knowledge and above all to secure for the people of the country all benefits that accrue from the acquisition and application of scientific knowledge. The focus of five-year plans on science and technology was on the needs of the times and continuously evolving.

The **First Plan** aimed at the setting up of new national laboratories and research institutes, training the personnel for manning the research institutes and running industries.

The **Second Plan** focused on generating scientific human resources insufficient.

The **Seventh Plan** recognized new thrust areas in S&T emerging on the world scene, such as microelectronics, informatics, telemetric, robotics, biotechnology, material science, instrumentation etc.

The **Eighth Plain** sought to integrate science and technology win socio-economic sectors and more particularly rural areas to meet the basic needs of water, nutrition, health and sanitation, shelter, energy, education and employment.

The **Ninth Plan** emphasized on self-reliance due to growing global restrictions on high-technology movement.

The **Eleventh Plan** envisaged the formulation of a National Innovation Policy to encourage competition among enterprise and greater diffusion of knowledge. Other proposals include the Council for Advancement of People's Action and Rural Technology (CAPART) to become an effective link between technology generation centres and various ministries for dissemination and propagation of technology packages through employment generation and capacity building schemes. Despite much progress, a very large part of the Indian population continued to be excluded from the benefits of S&T applications.

The **Twelfth Plan** aimed to overcome this exclusive if the goal of inclusive development in the Decade of innovation has to be me. It dealt with inclusive health; biodiversity-based knowledge systems and grassroots innovations; linking modern S&T with traditional arts and other artisanal goods, sustainable land use and climate change etc.

SCIENCE AND TECHNOLOGY UNDER NITI AAYOG

NITI Aayog replacing erstwhile planning commission continues to focus on technology-driven growth by proceeding technology up-gradation and capacity building for implementation of programmes and initiatives.

NITI Aayog taken the initiative on Blockchain usages in E-governance and has conceptualized the tech stack as "India Chain". India Chain is the name given to Niti Aayog's ambitious project to develop a nation-wide blockchain network. The vision is to link India Chain with India Stack, the digital infrastructure that forms the backbone of the Aadhar project. The NITI Aayog initiative on the blockchain system will enforce contracts quicker, prevent fraudulent transactions. And help farmers through the efficient disbursement of subsidies. This project is the first step to a larger system of record keeping and public good disbursement.

Atal Innovation Mission (AIM) including Self-Employment and Talent Utilization (SETU) under NITI Aayog is Government of India's endeavor to promote a culture of innovation and entrepreneurship. Its objective is to serve as a platform for the promotion of world-class innovation Hubs, Grand Challenges, Start-up businesses and other self-employment activities, particularly in technology driven areas.

SCIENCE AND TECHNOLOGY APPLICATIONS AND EFFECTS IN EVERYDAY LIFE

Introduction:

In present global scenario, application of science and technology is increasing at great pace. Global populace depends on technology and they use various technologies to accomplish specific tasks in their lives. Today, people have various emerging technologies which impact their lives in different ways. Technology is being implemented in almost every section of lives and business structures.

Therefore, it is important to embrace it and learn how to use technology in business and lifestyle. Due to high speed development and rapid changes in world, technology will be changed. So, it is better to go with latest trends and new emerging technologies and learn how to embrace and use them in daily life.

Use of technology in education:

Modern technology has made major change in the education field. With the invention of technological gadgets and mobile apps which helps students to learn easily. Now-a-days students can access a full library via a mobile app on any smart phone or ipad. Before inventing this technology, students had to go to physical libraries to get the information they need. Teachers can also use modern technology to teach their students.

The Government of India has initiated major reforms in the education sector to improve the quality and access to education. Aimed at expanding access to quality education, new premier higher education institutions have been opened across the country, marking the highest ever such expansion in the history after Independence.

1. **SWAYAM** MOOCs portal (Study Webs of Active learning for Young Aspiring Minds) is an indigenous MOOCs portal that provides high quality education -

anyone, anytime, anywhere at no cost- has been made operational. The portal has courses by the best teachers in the country and offers video lectures, e-reading material, discussion forum and assessment system - resulting in award of credits to the successful learners. So far, more than 1000 courses are made available and more than 33 lakh users are registered in this forum. Upto 20 % credit transfer into the academic record of students, is allowed now, for the courses taken on the SWAYAM portal. To reach high quality educational content to the most backward areas using satellite communication, 32 DTH channels have been made functional, under **SWAYAM Prabha** programme. These channels air 4 hours of new content every day

2. The initiative of **National Digital Library (NDL)** is a virtual repository of learning resources with a single window search facility. It has already brought together 15 million digital books and journals and 31 lakh learners are using this facility. It supports all academic levels including researchers and life-long learners, all disciplines, all popular form of access devices and differently abled learners.
3. **e-pathshala** has been developed by NCERT (National Council for Educational Research and Training) for showcasing and disseminating all educational e-resources including textbooks, audio, video, periodicals and a variety of other print and non-print materials. So far, 3,062 audios and videos, 650 e-books (e-pubs) and 504 flip books have been made available on the portal and mobile app.
4. '**Operation Digital Board**' in all schools (class 9 onwards) and colleges as a joint effort of Centre, States, CSR and community initiatives to provide good quality education by effective use of technology and telecom services and convert every class room in the country (from class IX and above) into a digital class room.

Use of technology in banking:

Banking sector is advanced due to various technological developments. Now electronic banking is used to transfer money and it eases customers.

Bank net is a first national level network in India, which was commissioned in February 1991. It is communication network established by RBI on the basis of recommendation of the committee appointed by it under the chairmanship of the executive director T.N.A. Lyre.

E-banking: This enables the bank to deliver its services easily to its high-end customers. To make the system user friendly to all clients, banks have used a Graphical User Interface (GUI), with this software, customers can access their bank details on their own computers, make money transfers from one account to another, print bank statements and inquire about their financial transactions. Another technology used by banks to exchange data between the bank and clients is called Electronic Data Interchange (EDI); this software can be used to transmit business transaction in a computer-readable form. So, the client on the other end will be in position to read the information clearly.

NRI Banking Services: This technology has been embraced in countries like India, USA, UAE, just to mention but a few. Since many people go abroad to work, they have a need of supporting their families. So, technology has made it simple for them to send money to their loved ones easily.

Plastic money: Credit cards or smart cards like **"**VISA ELECTRON**"** have made the banking industry more flexible than before. With a credit card, a customer can borrow a specific amount of money from the bank to purchase any thing and the bank bills them later. In this case, they don't have to go through the hassle of borrowing small money. Then with ''Smart Cards" like visa electron, a customer can pay for anything using that card and that money is deducted from their bank accounts automatically, they can also use the same card to deposit or withdraw money from their accounts using an ATM machine.

Use of Technology in Agriculture:

Technology has played a big role in developing the agricultural industry. Today it is possible to grow crops in a desert by use of agricultural biotechnology. With this technology, plants have been engineered to survive in drought conditions. Through

genetic engineering scientists have managed to introduce traits into existing genes with a goal of making crops resistant to droughts and pests.

A bacterium known as ”Bacillus Thuringiensis” acts like a reservoir , it enables crops to be insect-resistant , so these genetically modified crops will grow without any interference from pests. The invention of this technology is being used in developing countries to grow cash crops like cotton, since this genetically engineered cotton plants are pest resistant, they grow better than the normal cotton plants hence yielding good results.

Genetically produced plants like potatoes, can resist diseases and pests, which rewards the farmer with good yields and saves them time. These crops grow very fast they produce healthy yields. Since they are resistant to most diseases and pests, the farmer will spend less money on pesticides, which in return increases on their (RIO) return on investment.

Questions:

1. Describe the role of the Science and Technology in India.
2. Mention the Achievements of India in the field of Science and Technology.
3. Highlight the Development of Science and Technology in India’s Pre independence Stage.